

Ruiyang Wang

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EDUCATION

California State Polytechnic University, Pomona

Bachelor of Science in Computer Science, Minor in Data Science

Expected May 2026

GPA 3.78

- *President's Honor List: 2023-2024, 2024-2025*
- *Dean's List: Fall 2023, Spring 2024, Fall 2024, Spring 2025*

Massachusetts Institute of Technology (Online)

Professional Certificate in Data Science and Machine Learning: Making Data-Driven Decisions

Completed Feb 2025

RELEVANT COURSEWORK

Artificial Intelligence • Generative AI • Web Search & Recommender Systems • Design and Analysis of Algorithms • Computer Architecture • Big Data Analytics and Cloud Computing

FELLOWSHIPS & GRANTS

- Engage Student Stipend (\$2,000), Cal Poly Pomona
Competitive scholarship supporting research

2025-2026

RESEARCH EXPERIENCE

Anomaly Detection Research Fellow - Cal Poly Pomona, Computer Science Department

Advisor: Dr. John Korah

Sep 2024 - Aug 2025

- Designed and implemented vertically partitioned deep learning architecture from scratch in Python for real-time network anomaly detection on CIC-IDS2017 dataset, enabling full control over distributed message-passing logic and layer structure
- Achieved 217× speedup through MPI-based parallelization across 32 processors; conducted scalability benchmarking (4-64 processors) demonstrating performance peaked at 32 before communication overhead dominated, validating Amdahl's Law predictions
- Validated numerical correctness against functionally equivalent PyTorch baseline through rigorous stability analysis of floating-point precision during distributed communication

Engage Research Scholar — Cal Poly Pomona, Engage Program

Advisor: Dr. Denis Iurchenko

Aug 2025-Present

- Investigating how generative AI linguistic features influence trust and decision-making in high stakes crowdfunding environments by analyzing large-scale datasets from Crowdcube and Republic platforms
- Examining relationship between AI-generated project updates, founder-investor communication patterns, and funding success rates to understand human adoption of generative AI tools
- Developing creator-focused evaluation frameworks for controllable generation systems beyond technical benchmarks (FID scores), incorporating constraint-satisfaction rates, iteration efficiency, and user preference metrics

PROJECT EXPERIENCE

Gen Z Slang Explainer - Cal Poly Pomona, Generative AI Course

Oct 2025 – Dec 2025

- Fine-tuned TinyLlama-1.1B (1.1B parameters) using LoRA on 1,700+ slang terms, achieving 80% format accuracy while demonstrating parameter-efficient transfer learning on unseen variations
- Designed evaluation protocol measuring both technical accuracy and contextual appropriateness of generated explanations

MIT Data Science & Machine Learning Capstone Projects (Online)

Nov 2024 – Feb 2025

- Amazon Recommendation System: Built collaborative filtering models (user-based and item-based) on 65,000+ ratings; evaluated using production metrics (Precision@k, Recall@k, NDCG)

- EdTech Lead Conversion: Developed predictive classifier optimized for business ROI; performed feature engineering and class imbalance handling achieving 0.76 ROC-AUC

*Energy Consumption Analysis — **Cal Poly Pomona Information Technology Competition 2025*** Apr 2025

- Directed team of 5 analyzing utility energy trends using Random Forest Regressor and exploratory data analysis to identify consumption patterns
- Presented data-driven cost-saving recommendations based on predictive usage forecasting

*Wildfire Alert Chatbot— **Bronco Hackathon 2025*** Apr 2025

- Led team of 4 developed and deployed a full-stack wildfire alert system integrating NASA FIRMS API for real-time satellite fire detection with Google Gemini API for natural language queries
- Built React frontend and Django backend deployed on Render with live geospatial data visualization

LEADERSHIP EXPERIENCE

*Chinese Students and Scholars Association (CSSA) - **Vice President - Cal Poly Pomona*** Jan 2024-Present

- Lead event planning and media coordination for 6-10 cultural festivals, academic workshops, and networking events per semester (50-200+ attendees)
- Developed Python QR code registration tool streamlining check-ins and data tracking for 300+ event attendees
- Serve as CSSA representative at Chinese Consulate (Los Angeles) events, coordinating visa guidance, academic resources, and community support for Chinese students
- Manage bilingual social media presence (Instagram, WeChat); create culturally accessible infographics and promotional materials

*LoopBreakers Competitive Technical Team - **Founder & Team Lead - Cal Poly Pomona*** Apr 2025 - Present

- Founded cross-disciplinary team for hackathons and technical competitions, lead project strategy, team coordination, and rapid prototyping initiatives across multiple competitions

WORK EXPERIENCE

*Data and Visa Processing Summer Intern - **Embassy of Japan - Beijing, China*** Jul 2023 - Aug 2023

- Improved verification workflow reducing cycle time by ~20%; processed 50-70 visa applications daily with zero rework using government database systems

SKILLS

Programming Languages: Python, Java, C, JavaScript, Kotlin, SQL

AI/ML Frameworks: PyTorch, scikit-learn, Transformers (Hugging Face), LoRA/PEFT, MPI (mpi4py), Pandas, NumPy

Development Tools: Linux, Git, Docker, Jupyter Notebook, Android Studio

Cloud & Visualization: Azure, Power BI, Tableau

Other Skills: UI/UX Design (HTML/CSS), Data Analysis, Digital & Traditional Illustration, Visual & Media Design, Canva

Language Skills: Bilingual in Chinese and English